

COOKNOOK: Intelligent Meal Planning Application

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ABSTRACT

In Canada, nearly 40% of university students experience food insecurity [1, 2]. However, most young adults and students are not aware or even underestimate the extent of the insecurity they face. This phenomenon can increase the cognitive burden associated with food management, leading to poor resource allocation exacerbating levels of food insecurity. This negatively impacts Sustainable Development Goal 2, 'Zero Hunger.' The Canadian government and university institutions have often relied on food banks or charitable organizations. However, these interventions have mainly served as emergency measures rather than long-term solutions [3]. Through research based on a design-adapted ethnographic approach, our team has gained a deep understanding of the situation of university students and the challenges related to food. Consequently, we propose CookNook: an application featuring an intelligent personal assistant aimed at helping users prioritize cooking time by reducing the cognitive load associated with food. This is accomplished by providing simple and accessible recipes, accompanied by shopping lists tailored to their resources, needs, and constraints. CookNook also facilitates culinary gatherings with friends to share meals while motivating them to cook more. In this article, we summarize our design process and how our solution represents a step towards ensuring sufficiently quality nutrition among university students.

CCS CONCEPTS

• **Human-centered computing**; • **Interaction design**; • **Interaction design process and methods**; • **User centered design**;

KEYWORDS

Food insecurity, University students, Artificial intelligence, Mobile application, Human-centered design

ACM Reference Format:

Myriam Hamada*, India Tanguay Doucet*, and Brissac Aurélie. 2024. COOKNOOK: Intelligent Meal Planning Application. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3647979>

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1 INTRODUCTION

1.1 Food Insecurity

According to Statistics Canada, food insecurity is the inability to acquire or consume an adequate quantity and quality of food in a socially acceptable manner, or uncertainty about where one's next meal will come from [4]. It takes three forms: marginal food insecurity, expressing concern about running out of food; moderate food insecurity, resulting in a reduction in the quality or quantity of food; and severe food insecurity, characterized by skipping meals, reducing food intake, or even going entire days without food in the most severe cases.

1.2 The Situation in Canada and Quebec

In Canada, 30.7% of students face moderate food insecurity, and 8.3% experience severe food insecurity [5], posing a significant challenge to **SDG 2, 'Zero Hunger.'** The high costs of food, tuition, and housing contribute to this situation [5]. The constant rise in educational, food, and housing costs, coupled with a significant 10.1% increase in food prices in Quebec in 2023 [6], impacts post-secondary students similarly to the general public. The substantial increase in tuition fees, almost doubling between 1993 and 2016, puts significant financial pressure [5]. The recent food inflation in Quebec further exacerbates the situation, forcing students to compromise on the quality, quantity, and diversity of their food purchases. Temporary solutions from the government and university institutions often address only a portion of the student population. It becomes crucial to find sustainable solutions to ensure regular, affordable, and quality nutrition for university students.

1.3 Support for Accessible and Quality Nutrition

For over three months, we adopted a human-centered design approach by conducting design-adapted ethnographic research within the student community at Laval University in Quebec City. After observing the eating habits of students, we identified a lack of time as a major obstacle, attributable to a hectic lifestyle aimed at high academic performance. The primary factors contributing to this phenomenon include the perceived high cognitive load associated with meal planning, cooking, and overall mental well-being. Underlying reasons often masked by claims of time constraints lead students to rely more on fast-food, cafeteria options, or pre-prepared meals, exacerbating resource mismanagement and, consequently, escalating levels of food insecurity. Based on our findings and an iterative design process, we present CookNook: an application with an intelligent personal assistant offering simple recipes and shopping lists tailored to users' resources, needs, and constraints. Furthermore, CookNook facilitates social culinary events with friends, fostering a supportive environment that promotes cooking motivation through

shared experiences and enjoyable moments. This article explores existing solutions, emphasizing the need for assistance specifically addressing students' needs, and then details our user research and design process, presenting the conceptual version of CookNook.

2 CITING RELATED WORK

Food assistance organizations or low-cost grocery stores are available to students, but often their accessibility is limited. For example, at Laval University in Quebec, Canada, the food bank La Table du Pain [7] distributes food items to students free of charge. However, the quantity of products is limited, distributions occur only once a week during a two-hour time slot, and eligibility criteria are quite restrictive. In Quebec City, there are very low-cost grocery stores offering discounts for students. However, these stores, located on the outskirts of the city, are difficult or out of reach for most students.

Regarding technologies designed to facilitate access to food, applications such as Reebee [8] and Flipp [9], freely available on mobile download platforms, provide information about daily discounts and special offers in local grocery stores. Others, like Too good to go [10] and Sauvegarde [11], connect restaurants, grocery stores, and local retailers with surplus food to consumers, offering them affordable food items. Marmiton [12] along with many other applications, provides a variety of recipe ideas. However, applications providing information on discounts or surplus food often assume that the consumer has enough time and personal transportation to make their trips worthwhile. This is not always the case for students. Additionally, apps offering recipe ideas may not always be tailored to their needs and constraints. Consequently, the services offered by these applications do not specifically cater to university students and do not entirely meet their needs. These apps highlight the need to further explore how technology can assist them in accessing more affordable and high-quality nutrition.

3 USER RESEARCH AND METHODS

3.1 Graffiti Wall

The Graffiti Wall method is an open platform that allows participants to freely share written or visual comments on a phenomenon, promoting free and anonymous expression [13]. The use of this method is based on the principle of anonymous testimonials, without judgment, thus fostering greater freedom of expression, especially on sensitive topics such as food insufficiency. However, its limitations include limited control over participants and the difficulty of precisely identifying contributors. Despite this, it was successfully implemented for 14 days in a student café, collecting over thirty responses highlighting recurring themes such as lack of time, mental state, and the workload associated with studies.

3.2 Questionnaire

The questionnaire method is employed as a survey instrument specifically designed to collect self-reported information from individuals, focusing on their characteristics and thoughts [13]. This methodological choice was motivated by the inherent simplicity in its development and administration. Additionally, software and online services were identified as excellent resources for efficient construction and distribution of questionnaires. In our study, the

questionnaire was created using the *Google Form* tool, and we obtained a total of 25 responses. The questionnaire content included socio-ethnographic questions, as well as questions defined based on the emerging themes from the Graffiti Wall, covering dietary habits, mental state, and workload. The collected data were subsequently analyzed to deepen our understanding of the participants' behaviors and experiences.

3.3 Semi-Structured Interview

Interviews, as a research method, are used as a means of collecting data from students in various degrees of food insecurity [13]. This approach is favored for its ability to establish direct contact with participants, facilitating the collection of first-hand personal testimonials about their experiences, opinions, attitudes, and perceptions. In our study, the choice was semi-structured interviews, characterized by basic questions that could lead to other themes, due to their conversational advantage and increased comfort for participants. Our team conducted interviews with a sample of 11 individuals, with each interview lasting an average of 20 to 30 minutes. The main objective of this approach was to gather more personal testimonials about students' eating habits and practices. To achieve this, we defined our interview questions based on the emerging themes from the Graffiti Wall and the Questionnaire. The most recurrent testimonials addressed topics such as financial means, mental state, and influences from the social environment. This methodology allowed us to deepen our understanding of students' eating behaviors, highlighting essential aspects of their daily experience.

3.4 Data Analysis : affinity diagram

Following our user research through the Graffiti Wall, questionnaire, and interviews, we analyzed the collected data using an Affinity Diagram. The Affinity Diagram is a compilation of large amounts of data organized into groups or themes based on their relationships [14]. This analysis method was paired with the 'AEIOU framework' (Activities, Environment, Interactions, Objects, Users). According to the book 'Beyond Design and Ethnography' by Nicolas Nova [15], the 'AEIOU framework' involves categorizing data into 'Activities', 'Environment', 'Interactions', 'Objects', or 'User Typologies' to gain different perspectives on understanding the data. The affinity diagram process combined with the AEIOU framework proved particularly useful in our case to make sense of the large volume of mixed information and data collected during our qualitative research. From these analysis methods, the following themes emerged: the role of mental load in meal organization, eating habits and strategies, and social issues and mental state.

4 OBSERVATIONS AND FINDINGS

4.1 Organization and Mental Load

In both the questionnaires and interviews, many participants highlight the mental load and stress associated with meal organization, especially budgeting for groceries. The complexity of this step, demanding time, discipline, and sometimes challenging to adhere to due to a lack of time and motivation, is emphasized. Multiple

constraints, such as time pressure, budgetary restrictions, and managing already purchased food items, are expressed. Stock management, with difficulties in monitoring expiration dates and avoiding waste, poses a significant challenge. The level of food knowledge is essential in meal management. Overall, the mental load related to meal organization is a recurring theme, underscoring the need for innovative solutions to promote a less stressful meal management.

4.2 Eating Habits and Emergency Strategies

Following the questionnaires and interviews, two categories of strategies emerge for managing meals and participants' budgets. Positive strategies, favored with good management of time, money, and motivation, include creating grocery lists, budgets, meal prep, and lunch preparation. Recipe inspirations from social media, tailored to students, were also highlighted. On the other hand, emergency strategies, adopted in cases of time, money, or motivation constraints, involve purchasing lower-quality processed foods, such as canned or dehydrated foods, as well as buying ready-made meals, which are more expensive. These two categories of strategies highlight the practical choices and compromises that students face in the daily management of their nutrition, emphasizing the need to understand these different approaches.

4.3 Social Issues and Mental State

Most interviewed individuals emphasize the significant impact of their social environment on their eating practices, whether it be family, roommates, friends, or colleagues. Some receive direct assistance from their family, such as meal preparation, which is often beneficial. Co-habitation promotes mutual assistance and sharing of food supplies, less common among those living alone. Social outings have a financial impact, sometimes unforeseen. Mental state, including fatigue, stress, anxiety, and mood, influences motivation and commitment to meal planning. These findings underscore the importance of a holistic and contextualized approach to eating practices.

4.4 Design Opportunity

From these observations and findings, a design opportunity emerges for a product or service that could facilitate inventory management, offer personalized recipes considering users' time and budget constraints, and incorporate their level of food knowledge. Additionally, it could connect individuals for productive and cost-effective social events. This solution would present a clear value proposition overcoming the limitations of existing food-related applications. Unlike current food assistance services, this service or product would offer a flexible solution specifically tailored to the needs of university students.

5 DESIGN PROCESS

5.1 Ideation and Proposed Concept

Our team leveraged these findings to initiate an iterative ideation process, leading to several concepts. One of them considered creating a university food assistance program supported by an application, aiming to establish connections between university organizations, student cafés, food banks, and related study programs.

The goal was to offer food education and facilitate access to quality and affordable food for the student community. Another idea involved organizing surplus collection activities directly from farms or wholesalers using an application. While both concepts addressed the identified needs of students, they were dismissed due to the challenges associated with establishing partnerships with services and organizations not always easily accessible, as well as adding an additional activity to the students' schedules. Ultimately, to provide a more holistic and less constraining solution, the team focused on a personalized intelligent assistant in the form of a mobile application, aimed at assisting students in meal and grocery planning while facilitating food inventory management to avoid waste. This solution aligns with **Sustainable Development Goal (SDG) 2: 'Zero Hunger'** and partially contributes to **SDG 12: 'Responsible Consumption and Production'**.

5.2 First Iteration : Features

In the initial phase of our research, we proceeded with the comprehensive identification of features that the application would offer. To visualize this conceptualization better, we created a table inspired by the structure of a blueprint. This table was designed to provide a visual representation of the various phases and steps of using the application, correlated with corresponding features, devices involved, and the underlying technological architecture of the process. The chronological layout of the table allows for a sequenced understanding of interactions and components, providing a clear overview of how the application would be used.

5.3 Second Iteration : Form and Structure

After establishing the full set of application features, our team undertook a second phase involving the creation of low-fidelity prototypes. These prototypes, developed in the form of wireframes (fig. 1) using the Figma tool, helped group and organize the various features into several sections of the application, defined as distinct interfaces. The aim of this approach was to prepare user tests to ensure that the application is perceived as useful and easy to navigate.

5.4 User Testing and Corrections

Once the form and structure of the application were defined, and the wireframes were created, our team implemented a participatory evaluation method involving low-fidelity interactive prototype testing [16]. The aim of this method was to check the ease and intuitiveness of navigation, identify potential pain points, and gather new ideas. To prepare for the tests, we prototyped the wireframes following pre-established user journeys, such as "account creation" or "meal menu generation" (fig. 1). We defined user goals in the form of tasks. The tests, conducted using the Figma tool on mobile and tablet devices, lasted approximately 20 to 30 minutes and were performed by a total of four testers. Testers were required to complete assigned tasks while commenting on their actions, thoughts, and hesitations aloud. The test results, in the form of notes and video sequences of screen captures, allowed our team to make adjustments such as changing labels, reviewing the layout of certain clickable elements, reorganizing sections of the application, and smoothing certain navigation paths.

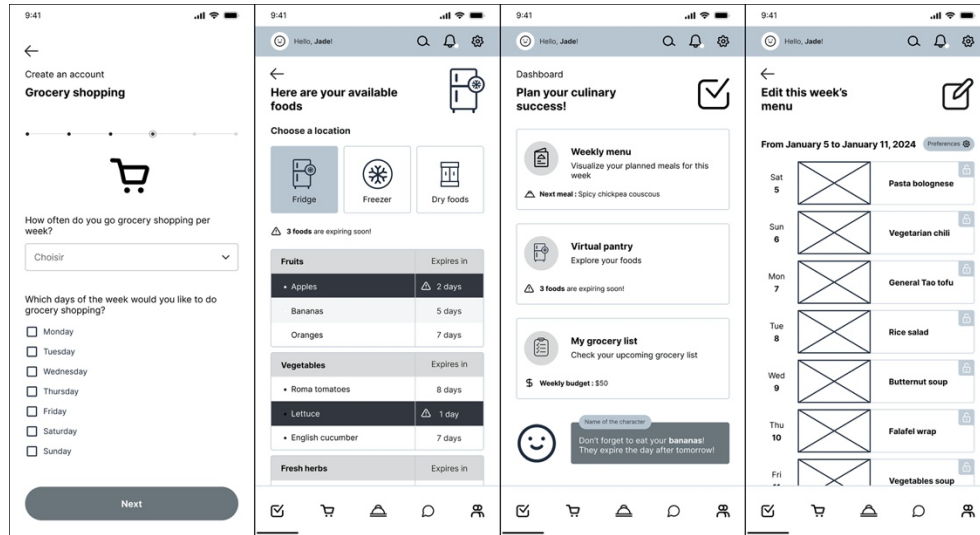


Figure 1: Low-fidelity prototypes. Wireframes of 4 interfaces: account creation, pantry, dashboard, and menu.

5.5 Focus Group

A focus group consisting of ten university students was organized to assess the level of acceptance of the concept. The students raised certain questions and limitations that allowed us to improve our solution. One initial question concerned the possibility of connecting multiple accounts to the same bulletin board or creating a shared account for roommates. Another question was about the frequency of meal planning. This prompted us to personalize the onboarding process, considering preferences for food purchases and meal preparation. Some participants questioned their level of commitment to the meal plan. They suggested adding notifications to track recipe completion, while being mindful not to impose unnecessary pressure. One student asked if the application would be useful to her even if she isn't constrained by financial or time limitations. In response, we emphasized the community aspect of the application, gamification, and the ability to explore more challenging recipes to enhance culinary skills.

6 FINAL ITERATION: COOKNOOK

The final version of the application incorporated all the feedback and observations obtained from previous cycles to make it effective and user-friendly. A visual identity was created and applied to the interfaces to provide a professional and realistic look to the renders. Our team presents CookNook, a multi-device application with an AI-powered assistant, Nook. CookNook offers students personalized, simple, and accessible recipes, accompanied by shopping lists tailored to their resources, needs, and constraints to better manage their food resources and reduce waste. To showcase this final version effectively, the concept presentation was done chronologically and divided into three themes: customization and automation, meal planning and adapted recipes, and connection and sharing. CookNook primarily operates on mobile devices, offering users a convenient and accessible platform.

6.1 Customization and Automation

In the initial phase, the student creates their profile, answering questions about their eating habits, budget, and grocery logistics (fig. 2). After that, they can input their schedule either manually or automatically through the application's access to their digital calendar (Google Calendar, iCloud Calendar). These data are saved for future use. The "guided tour" directs the student to the main sections of the application such as the "weekly menu," "pantry," and "grocery list" on the dashboard (fig. 2). Secondary sections like "kitchen," "community," and "events" are accessible from the navigation bar. Throughout the onboarding process, students complete the "pantry" section by inputting details about their food inventory, including items stored in the refrigerator, dry goods, and freezer. This can be done manually or using multiple-object image recognition technology. While setting up the account and pantry may appear time-consuming, it is a crucial step that occurs only during the initial installation of CookNook. Fortunately, the integration of multiple-object recognition technology significantly simplifies these tasks.

6.2 Meal Planning and Adapted Recipes

Once the stocks are recorded, the application associates this data with eating habits to create a personalized menu in the "Weekly Menu" section (fig. 2). The menu generation, based on personal data such as budget, stocks, and available time, uses online databases to access recipes via web scraping [17]. The menu, entirely customized based on food preferences, budget, and stocks, can be modified by the student. Once validated, the application generates a minimal and economical shopping list, taking existing stocks into account. The student uses this list for their purchases, automatically updating the virtual pantry as they check items or scan their receipt. This automation ensures up-to-date stocks and facilitates the creation of updated meal plans, establishing an automated virtuous circle.

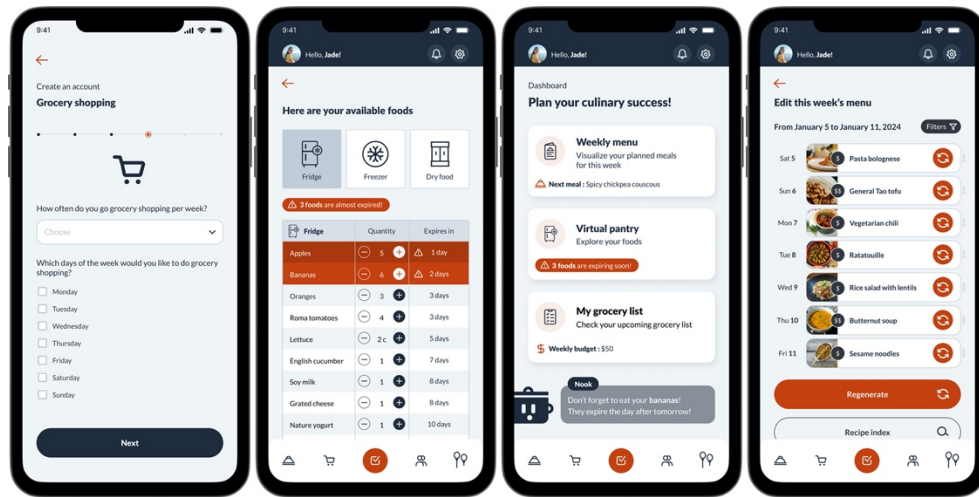


Figure 2: High-fidelity prototypes. Wireframes of 4 interfaces: account creation, pantry, dashboard, and menu.

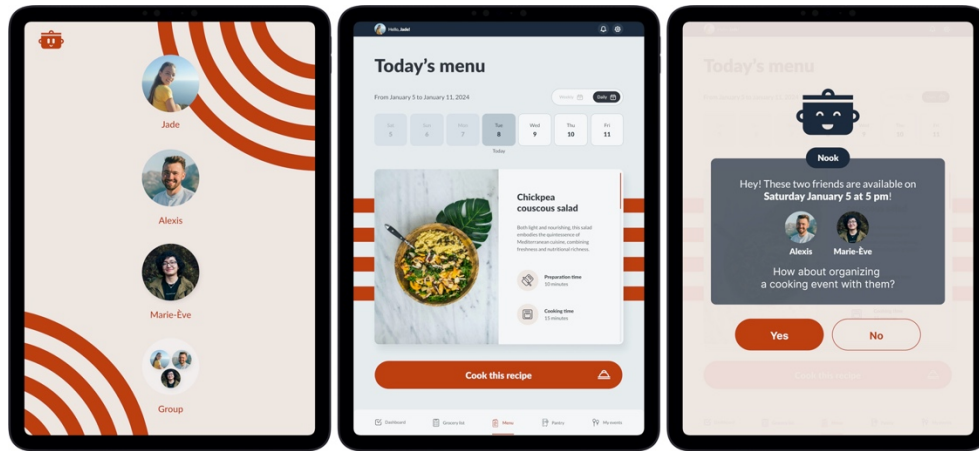


Figure 3: High-fidelity prototypes of the bulletin board. Wireframes of 3 interfaces: choosing an account, today's menu and notification suggesting a cooking event.

Finally, the "kitchen" section offers step-by-step assistance for dish preparation, enhancing the user's culinary experience.

6.3 Networking and Sharing

CookNook offers a relevant feature: the creation of events among friends. Students need to have friends using the app to benefit from this feature. The system uses users' schedules and pantries to suggest culinary events. It only shows common time slots, to preserve privacy of one's personal calendar. This function promotes cost-effective social activities by enabling food sharing. Events can be random or requested by users. The "community" section facilitates sharing, and gamification processes encourage participation. For example, the "International Chef" badge is obtained by making recipes from all continents.

6.4 Bulletin Board

Additionally, users have the option to obtain an electronic bulletin board as a supplementary feature (fig. 3). This bulletin board aids in visualizing pantry management and provides step-by-step guidance during recipe preparation. Users can strategically place the tablet for easy access while cooking, facilitating a seamless cooking experience. It's important to highlight that while the electronic bulletin board is available, it is entirely optional and considered a "nice-to-have" addition. We recognize that some students may have budget constraints that prevent them from acquiring the tablet, and thus, CookNook functions effectively without it, ensuring accessibility for all users.

7 CONCLUSION

In conclusion, CookNook innovates by simplifying students' food management through personalized recipes and the automation

of mentally demanding tasks. The virtual assistant, Nook, adds a friendly dimension to the experience. By promoting social connections and culinary events among friends, the application transforms the routine of eating into a collaborative activity. This collective approach encourages users to prioritize cooking affordable dishes, ultimately reducing expenses linked to pre-made meals or fast food. An extension for managing shared food supplies in shared living arrangements could be promising to meet the specific needs of students in a community setting. A partnership with universities could facilitate the application's dissemination to new students. With adjustments based on feedback, CookNook could broaden its impact to include young adults and families, highlighting its evolutionary potential and positive impact.

ACKNOWLEDGMENTS

In acknowledging the partial implementation of the solution, we note the inherent limitations of CookNook, including the complexity of its technological architecture, which encompasses specific details of artificial intelligence and underlying programming to automate complex tasks, not explicitly discussed here. The inability to conduct concrete tests hinders the validation of the actual capabilities of artificial intelligence. However, technology is constantly evolving, providing prospects for future improvement. While full applicability may be deferred, technological progress opens opportunities for enhancing and expanding CookNook's features.

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